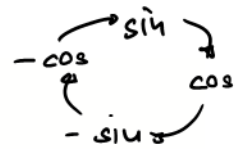


Analysis V Aufgaben Dario Hug



V. Weiterführung Differentialrechnung

Trigonometrische Funktionen:

6. Leite zweimal ab.

a) $f(x) = 4 \cos(x)$ b) $f(x) = \sin(3x)$ c) $f(x) = \sin(1-x)$ d) $f(x) = \cos(3x - \pi)$

e) $f(x) = \sin(3x) - \cos(x - \pi)$ f) $f(x) = \sin(2x) + \sin(x)$

a) $f'(x) = -4 \sin(x) \rightarrow f''(x) = -4 \cos(x)$

b) $f'(x) = 3 \cdot \cos(3x) \rightarrow f''(x) = -9 \sin(3x)$

c) $f'(x) = \cos(1-x) \rightarrow f''(x) = -\sin(1-x)$

d) $f'(x) = 3 \sin(3x - \pi) \rightarrow f''(x) = 9 \cos(3x - \pi)$

e) $f'(x) = 3 \cos(3x) - \sin(x - \pi) \rightarrow f''(x) = -9 \sin(3x) - \cos(x - \pi)$

f) $f'(x) = 2 \cos(2x) + \cos(x) \rightarrow f''(x) = -2 \sin(2x) - \sin(x)$

7. Leite zweimal ab. Bestimme die Periode der Funktion $f(x)$. Untersuche den Graphen auf Nullstellen, Extrem- und Wendepunkte. Zeichne den Graph.

a) $f(x) = \sin(4x)$

(b) $f(x) = 3 \cos(0.5x)$

a) $f'(x) = 4 \cos(4x)$ $f''(x) = -16 \sin(4x)$

Nullstellen $\rightarrow$ Unendliche im Abstand von $\frac{\pi}{4}$

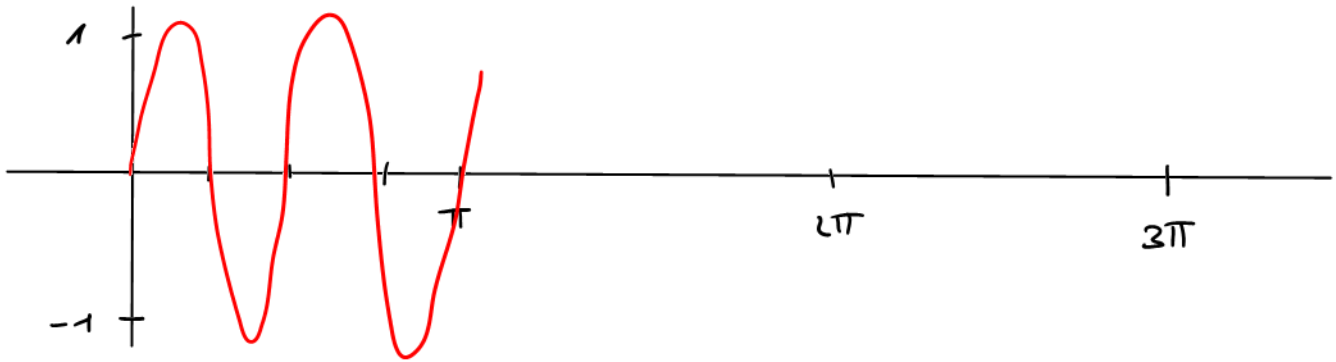
$\rightarrow x_n = k \cdot \frac{\pi}{4} \rightarrow k \in \mathbb{Z}$

NS = Wendepunkte

$\rightarrow x_w = k \cdot \frac{\pi}{4} \rightarrow k \in \mathbb{Z}$

Extremstellen im selben Absatz, jedoch versetzt

$$x_E = k \cdot \frac{\pi}{4} + \frac{\pi}{8} \rightarrow k \in \mathbb{Z}$$



b) $f(x) = 3 \cos(\frac{1}{2}x)$

$$f'(x) = -\frac{3}{2} \sin(\frac{1}{2}x) \quad f''(x) = -\frac{3}{4} \cos(\frac{1}{2}x)$$

$$x_N = 2\pi \cdot k \rightarrow k \in \mathbb{Z} (= x_w)$$

$$x_E = \frac{\pi}{2} + 2\pi \cdot k \rightarrow k \in \mathbb{Z}$$

8. Welche Kurve mit einer Gleichung der Form $K: y = a \cdot \sin(bx)$ hat die Periode $\pi/2$ und im Ursprung die Steigung 2?

Periode in Normalfall 2π

$\hookrightarrow \underline{b=4}$



$$a \cdot \sin(bx-c) \rightarrow a \cdot b \cdot \cos(bx-c)$$

$$f'(x) = 4a \cdot \cos(4x)$$

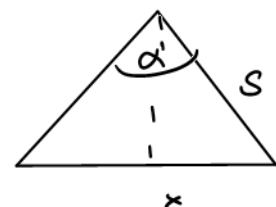
$$4a \cdot \cos(0) = 2$$

$$a = \frac{1}{2}$$

$$f(x) = \frac{1}{2} \cdot \sin(4x)$$

9. Wie ist bei einem gleichschenkligen Dreieck mit der Schenkellänge s der Basiswinkel α zu wählen, damit der Flächeninhalt des Dreiecks maximal wird?

$$A = \underbrace{\left(2 \cdot \sin\left(\frac{\alpha}{2}\right) \cdot s\right)}_x \cdot \left(s \cdot \cos\left(\frac{\alpha}{2}\right)\right) \cdot \frac{1}{2} \quad \begin{matrix} GAGA \\ HHAG \end{matrix}$$



$$f(\alpha) = s^2 \left(2 \cdot \sin\left(\frac{\alpha}{2}\right) \right) \left(\cos\left(\frac{\alpha}{2}\right) \right) \cdot \frac{1}{2}$$

$$f(\alpha) = 2s^2 \cdot \sin\left(\frac{\alpha}{2}\right) \cdot \cos\left(\frac{\alpha}{2}\right) \cdot \frac{1}{2} \rightarrow \sin(2\alpha) = 2 \cdot \sin(\alpha) \cdot \cos(\alpha)$$

$$f(\alpha) = s^2 \cdot \frac{1}{2} \cdot \sin(2\alpha)$$

$$f'(\alpha) = s^2 \cdot \cos(2\alpha)$$

$$s^2 \cdot \cos(2\alpha) = 0$$

$$s \neq 0$$

$$\cos(2\alpha) = 0$$

$$\alpha : 0 < \alpha < 180^\circ$$

$$\cos(90) = 0$$

$$\alpha = \frac{90}{2}$$

$$\alpha = 45^\circ$$

39.1 Die Produktregel:

Aufgabe: Vergleiche das Produkt der Ableitungen von $u(x) = x^2$ und $v(x) = x^3$ mit der Ableitung ihres Produkts.

4. a) Leite die Funktion $f(x) = (3x^2 - x)(x^3 - 1)$ nach der Produktregel ab.
b) Multipliziere erst die Klammern aus und leite danach ohne Produktregel ab. Vergleiche.
8. a) $f(x) = x \cdot \cos(x)$ b) $f(x) = (x^2 + 1)\sin(x)$ c) $f(x) = \sqrt{x} \cdot \cos(x)$
d) $g(t) = \sin(t) \cdot \cos(t)$ e) $g(t) = \sin^2(t)$ f) $f(x) = \cos^2(x)$

LSG:

39.1 Die Produktregel:

Aufgabe: $u'(x) = 2x$, $v'(x) = 3x^2 \Rightarrow u' \cdot v' = 6x^3$ und $u \cdot v = x^5 \Rightarrow (u \cdot v)' = 5x^4$

4. a)
- $$\begin{array}{lcl} u(x) = 3x^2 - x & \Rightarrow & u'(x) = 6x - 1 \\ v(x) = x^3 - 1 & \Rightarrow & v'(x) = 3x^2 \end{array} \Rightarrow \left| \begin{array}{ccc} u' & v & + \\ (6x - 1) & (x^3 - 1) & + \\ u & v' & \\ (3x^2 - x) & (3x^2) & \end{array} \right|$$
- $$f'(x) = (6x - 1)(x^3 - 1) + (3x^2 - x)(3x^2) = 15x^4 - 4x^3 - 6x + 1$$
- b) $f(x) = 3x^5 - x^4 - 3x^2 + x \Rightarrow f'(x) = 15x^4 - 4x^3 - 6x + 1$
8. a) $f'(x) = 1 \cdot \cos(x) + x \cdot (-\sin(x)) = \cos(x) - x \sin(x)$
b) $f'(x) = 2x \sin(x) + (x^2 + 1) \cos(x)$
c) $f'(x) = \frac{1}{2}x^{-\frac{1}{2}} \cos(x) - x^{\frac{1}{2}} \sin(x)$
d) $g'(t) = \cos(t)\cos(t) - \sin(t)\sin(t) = \cos^2(t) - \sin^2(t) = \cos(2t) = \dots$
e) $g'(t) = \cos(t)\sin(t) + \sin(t)\cos(t) = 2 \sin(t)\cos(t) = \sin(2t)$ (Additionstheorem ;)
f) $f'(x) = -\sin(x)\cos(x) - \cos(x)\sin(x) = -2 \sin(x)\cos(x) = -\sin(2x)$

39.2 Quotientenregel:

27. Leite zweimal ab:

$$-\rightarrow \left(\frac{u' \cdot v - u \cdot v'}{v^2} \right)$$

a) $f(x) = \frac{3}{x^2}$ b) $f(x) = \frac{2x+1}{x^2}$ c) $f(x) = \frac{x^2-x}{x+1}$ d) $f(x) = \frac{x^3}{1+x^3}$
 d) $f(x) = \frac{1}{\cos(x)}$ f) $f(x) = \frac{\cos(x)}{x}$ g) $f(x) = \frac{\cos(x)}{\sqrt{x}}$ h) $f(x) = \frac{\cos(x)}{\sin(x)}$

a) $f(x) = \frac{3}{x^2}$ $u(x) = 3$ $u'(x) = 0$
 $v(x) = x^2$ $v'(x) = 2x$

$$\frac{u' \cdot v - u \cdot v'}{v^2} \rightarrow \frac{0 \cdot x^2 - 3 \cdot 2x}{x^4} \rightarrow \frac{6x}{x^4}$$

$f'(x) = \frac{6}{x^3}$ $\rightarrow u(x) = 6$ $\rightarrow u'(x) = 0$
 $v(x) = x^3$ $v'(x) = 3x^2$

$$\rightarrow \frac{0 \cdot x^3 - 6 \cdot 3x^2}{x^6} \rightarrow \frac{-18x^2}{x^6} \rightarrow \underline{\underline{f''(x) = \frac{-18}{x^4}}}$$

b) $f(x) = \frac{2x+1}{x^2}$ $u'(x) = 2$ $\rightarrow \frac{2(x^2) - (2x+1)(2x)}{x^2}$
 $v'(x) = 2x$

$$\rightarrow \frac{2x^2 - 4x^2 - 2x}{x^4} \rightarrow \underline{\underline{f'(x) = \frac{-2x-2}{x^3}}}$$

$u(x) = -2x-2$ $u'(x) = -2$
 $v(x) = x^3$ $v'(x) = 3x^2$

$$f''(x) = \frac{-2x^3 - (-2x-2)3x^2}{x^6}$$

$$f''(x) = \frac{-2x^3 + 6x^3 + 6x^2}{x^6}$$

$$\underline{\underline{f''(x) = \frac{4x+6}{x^4}}}$$

$$c) f(x) = \frac{x^2 - x}{x + 1}$$

$$u(x) = x^2 - x \quad u'(x) = 2x - 1$$

$$v(x) = x + 1 \quad v'(x) = 1$$

$$\rightarrow f'(x) = \frac{(2x-1)(x+1) - (x^2-x)}{(x+1)^2}$$

$$= \frac{2x^2 + x - 1 - x^2 + x}{x^2 + 2x + 1}$$

$$\underline{\underline{f'(x) = \frac{x^2 + 2x - 1}{x^2 + 2x + 1}}}$$

$$\rightarrow u = x^2 + 2x - 1 \quad u' = 2x + 2$$

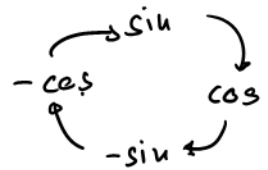
$$v = x^2 + 2x + 1 \quad v' = 2x + 2$$

$$\underline{\underline{f''(x) = \frac{(2x+2)(x^2+2x+1) - (x^2+2x-1)(2x+2)}{(x^2+2x+1)^2}}}$$

$$d) f(x) = \frac{1}{\cos(x)}$$

$$u = 1 \quad u' = 0$$

$$v = \cos(x) \quad v' = -\sin(x)$$



$$\underline{\underline{f'(x) = \frac{\sin(x)}{\cos^2(x)}}}$$

$$u = \sin(x) \quad u' = \cos(x)$$

$$v = \cos^2(x) \quad v' = \text{Produktregel!}$$

$$\hookrightarrow \cos(x) \cdot \cos(x)$$

$$\rightarrow f'(x) = u' \cdot v + u \cdot v'$$

$$\rightarrow u(x) = \sin(x) = -\sin'(x)$$

$$v(x) = \cos^2(x) = -\sin(x)$$

$$\rightarrow -\sin(x) \cdot \cos(x) + \cos(x) \cdot -\sin(x)$$

$$v'(x) = \hookrightarrow \underline{\underline{2(\cos(x) \cdot -\sin(x))}}$$

$$f''(x) = \frac{\cos(x) \cdot \cos^2(x) - (\sin(x) \cdot 2\cos(x) \cdot -\sin(x))}{\cos^4(x)}$$

$$= \frac{\cos(x) \cdot \cos(x) + 2\sin^2(x)}{\cos^3(x)}$$

$$\rightarrow \cos^2(x) + \sin^2(x) = 1$$

$$f''(x) = \frac{1 + \sin^2(x)}{\cos(x)}$$

$$f) f(x) = \frac{\cos(x)}{x}$$

$$u(x) = \cos(x) \quad u'(x) = -\sin(x)$$

$$v(x) = x \quad v'(x) = 1$$

$$f'(x) = \frac{u'v - uv'}{v^2}$$

$$f'(x) = \frac{-\sin(x) \cdot x - \cos(x)}{x^2}$$

$$\rightarrow u(x) = -x \sin(x) - \cos(x)$$

$$v(x) = x^2$$

$$u'(x) =$$

$$v'(x) = 2$$

$$\rightarrow -x \cdot \sin(x)$$

$$\rightarrow u(x) = -x \quad u'(x) = -1$$

$$v(x) = \sin(x) \quad v'(x) = \cos(x)$$

$$u' \cdot v + u \cdot v' \rightarrow$$

$$-1 \cdot \sin(x) + -x \cdot \cos(x)$$

$$\rightarrow \frac{u'v - uv'}{v^2}$$

$$u'(x) \rightarrow -\sin(x) - x \cdot \cos(x) + \sin(x) = -x \cdot \cos(x)$$

$$\rightarrow \frac{(-x \cdot \cos(x) \cdot x^2) - ((-x \cdot \sin(x) - \cos(x)) \cdot 2)}{x^4}$$

$$x \left((-\cos(x) \cdot x^2) - ((-\sin(x) - \cos(x)) \cdot 2) \right)$$

$$f'''(x) = \frac{-x^2 \cos(x) + 2x \sin(x) + 2 \cos(x)}{x^3}$$

$$x^4$$

$$f''(x) = \frac{(-\cos(x) \cdot x^2) + 2 \sin(x) + 2 \cos(x)}{x^3}$$

$$g) f(x) = \frac{\cos(x)}{\sqrt{x}} \quad \rightarrow u(x) = \cos(x) \quad u'(x) = -\sin(x)$$

$$\quad \quad \quad \rightarrow v(x) = x^{\frac{1}{2}} \quad v'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(x) = \frac{(-\sin(x) \cdot x^{\frac{1}{2}}) - (\cos(x) \cdot \frac{1}{2\sqrt{x}})}{x}$$

$$\rightarrow u(x) = (-\sin(x) \cdot x^{\frac{1}{2}}) - (\cos(x) \cdot \frac{1}{2\sqrt{x}}) \quad u'(x) =$$

$$v(x) = x \quad \rightarrow v'(x) = 1$$

$$a(x) = -\sin(x) \cdot \sqrt{x} \quad u(x) = -\sin(x) \quad u'(x) = \cos(x)$$

$$v(x) = \sqrt{x} \quad v'(x) = \frac{1}{2\sqrt{x}}$$

$$a'(x) = \cos(x) \cdot \sqrt{x} - \sin(x) \cdot \frac{1}{2\sqrt{x}}$$

$$a'(x) = \cos(x) \sqrt{x} - \frac{\sin(x)}{2\sqrt{x}}$$

$$b(x) = \cos(x) \cdot \frac{1}{2\sqrt{x}}$$

$$u(x) = \cos(x) \quad u'(x) = -\sin(x)$$

$$v(x) = \frac{1}{2\sqrt{x}} \quad v'(x) = c'(x) = \frac{1}{\sqrt{x}}$$

$$c(x) = \frac{1}{2\sqrt{x}}$$

$$u(x) = 1 \quad u'(x) = 0$$

$$v(x) = 2\sqrt{x} \quad u'(x) = 2 \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{x}}$$

$$c'(x) = 1 \cdot \frac{1}{\sqrt{x}} = \frac{1}{\sqrt{x}}$$

$$b'(x) = -\sin(x) \cdot \frac{1}{2\sqrt{x}} + \cos(x) \cdot \frac{1}{\sqrt{x}}$$

$$b'(x) = \frac{-\sin(x) + 2 \cos(x)}{2\sqrt{x}}$$

$$u'(x) = \cos(x) \sqrt{x} - \frac{\sin(x)}{2\sqrt{x}} + \frac{-\sin(x) + 2 \cos(x)}{2\sqrt{x}}$$

$$u'(x) = \frac{2\sqrt{x} \cdot \cos(x) \cdot \sqrt{x} - 2\sin(x) + 2\cos(x)}{2\sqrt{x}}$$

$$f''(x) = \frac{\frac{2\sqrt{x} \cdot \cos(x) \cdot \sqrt{x} - 2\sin(x) + 2\cos(x)}{2\sqrt{x}} \cdot x - (-\sin(x) \cdot x^{\frac{1}{2}}) - (\cos(x) \cdot \frac{1}{2\sqrt{x}})}{x^2}$$

$$g) f(x) = \frac{\cos(x)}{\sqrt{x}} \quad u(x) = \cos(x) \quad u'(x) = -\sin(x) \\ v(x) = \sqrt{x} \quad v'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(x) = \frac{-\sin(x) \cdot \sqrt{x} - \cos(x) \cdot \frac{1}{2\sqrt{x}}}{x}$$

$$f'(x) = \frac{-\sin(x) \cdot \sqrt{x} - \frac{\cos(x)}{2\sqrt{x}}}{x}$$

$$u(x) = -\sin(x) \cdot \sqrt{x}$$

$$v(x) = x \quad v'(x) = 1$$

$$u'(x): \quad a(x) = -\sin \cdot \sqrt{x} \rightarrow u(x) = -\sin \quad u'(x) = \cos(x) \\ v(x) = \sqrt{x} \quad v'(x) = \frac{1}{2\sqrt{x}}$$

$$a'(x) = -\cos(x) \cdot \sqrt{x} + -\sin(x) \cdot \frac{1}{2\sqrt{x}}$$

$$a'(x) = -\cos(x) \cdot \sqrt{x} - \frac{\sin(x)}{2\sqrt{x}}$$

$$b(x) = \frac{\cos(x)}{2\sqrt{x}} \rightarrow u(x) = \cos(x) \quad u'(x) = -\sin(x) \\ \rightarrow v(x) = 2\sqrt{x} \quad v'(x) = \frac{1}{\sqrt{x}}$$

$$b'(x) = \frac{-\sin(x) \cdot 2\sqrt{x} - \cos(x) \cdot \frac{1}{\sqrt{x}}}{4x}$$

$$b'(x) = \frac{-\sin(x) \cdot 2\sqrt{x} - \frac{\cos(x)}{\sqrt{x}}}{4x}$$

$$u'(x) = a'(x) - b'(x)$$

$$= \left(-\cos(x) \cdot \sqrt{x} - \frac{\sin(x)}{2\sqrt{x}} \right) - \left(\frac{-\sin(x) \cdot 2\sqrt{x} - \frac{\cos(x)}{\sqrt{x}}}{4x} \right)$$

$$= \frac{-4x \cdot \cos(x) \cdot \sqrt{x} - \frac{\sin(x) \cdot 4x}{2\sqrt{x}} + \sin(x) \cdot 2\sqrt{x} + \frac{\cos(x)}{\sqrt{x}}}{4x} \cdot 2\sqrt{x}$$

$$\frac{-8x^2 \cdot \cos(x) - 4x \cdot \sin(x) + 4x \cdot \sin(x) + 2\cos(x)}{8x^{\frac{3}{2}}}$$

$$u'(x) = \frac{-4x^2 \cdot \cos(x) + \cos(x)}{4x^{\frac{3}{2}}}$$

$$f''(x) = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{v(x)^2}$$

$$f''(x) = \frac{\left(\frac{-4x^2 \cdot \cos(x) + \cos(x)}{4x^{\frac{3}{2}}} \cdot x \right) - \left(-\sin(x) \cdot \sqrt{x} \right)}{x^2}$$

$$= \frac{-4x^3 \cdot \cos(x) + \cos(x) + \sin(x) \cdot \sqrt{x} \cdot 4x^{\frac{3}{2}}}{4x^{\frac{7}{2}}}$$

$$f''(x) = \frac{-4 \cos(x)x^2 + 4 \sin(x)x + 3 \cos(x)}{4x^{2.5}}$$

42 vollständige Funktionsuntersuchung (S.177ff)

6. Untersuche folgende Funktionen auf Symmetrie, Nullstellen, Extrema, Wendepunkte, Asymptoten. Zeichne anschliessend den Graphen.

a) $f(x) = \frac{3(2x-3)}{(x-3)^2}$

b) $f(x) = \frac{x^2 + 3x + 2}{x^2 - 2x - 3}$

c) $f(x) = \frac{x^3 - 8}{4x}$

d) $f(x) = \left(\frac{x-1}{x+1}\right)^2$

a) $f(x) = \frac{3(2x-3)}{(x-3)^2}$

→ 5 Eigenschaften

① Nullstellen:

$$3(2x-3) = 0$$

$$x = \frac{3}{2} \rightarrow \in \mathbb{D}$$

② Extremstellen

$$f(x) = \frac{3(2x-3)}{(x-3)^2}$$

$$u(x) = 3(2x-3) \quad u'(x) = 6$$

$$v(x) = (x-3)^2 \quad v'(x) = 2x-6$$

$$f'(x) = \frac{u'v - uv'}{v^2}$$

$$f'(x) = \frac{6(x-3)^2 - 3(2x-3)(2x-6)}{(x-3)^4}$$

$$\frac{-6x}{(x-3)^3} = 0$$

$$x = 0 \rightarrow \in \mathbb{D} \text{ \& Extremstelle}$$

$$f'(x) = \frac{6(x-3) - 6(2x-3)}{(x-3)^3}$$

$$f'(1) = +\frac{3}{4}$$

$$f'(-1) = -\frac{3}{32}$$

→ Lokales Minimum

$$f'(x) = \frac{-6x}{(x-3)^3}$$

③ Wendepunkte

$$f'(x) = \frac{-6x}{(x-3)^3}$$

$$u(x) = -6x \quad u'(x) = -6$$

$$v(x) = (x-3)^3 \quad v'(x) = 3x^2 - 18x + 27$$

$$f''(x) = \frac{(-6(x-3)^3) - (-6x(3x^2 - 18x + 27))}{(x-3)^6}$$

$$\frac{(-6(x-3)^3) - (-18x(x^2 - 6x + 9))}{(x-3)^6}$$

$$\frac{(-6(x-3)^3) - (-18x(x-3)^2)}{(x-3)^6}$$

$$\frac{-6(x-3) + 18x}{(x-3)^4}$$

$$\rightarrow \text{WP: } -6x + 18 + 18x = 0$$

$$\underline{\underline{x = -\frac{3}{2}}}$$

④ Symmetrie / Verhalten am Rand ($x \rightarrow \pm \infty$)

$\rightarrow$ Keine offensichtliche Symmetrie

$$\rightarrow \frac{6x-9}{x^2-6x+9} \rightarrow \frac{x(6-\frac{9}{x})}{x(x-6+\frac{9}{x})}$$

$$\rightarrow \frac{6-\frac{9}{x}}{x-6+\frac{9}{x}} \quad \lim_{x \rightarrow \pm \infty} \rightarrow \underline{\underline{0}}$$

⑤ Asymptoten

$$\rightarrow (x-3)^2 = 0$$

$x = 3$ $\rightarrow$ Vertikale Asymptote

$$b) f(x) = \frac{x^2 + 3x + 2}{x^2 - 2x - 3}$$

① Definitionsbereich:

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0 \quad \rightarrow \underline{\underline{\mathbb{D} = \mathbb{R} \setminus \{-1; 3\}}}$$

$$x_1 = 3$$

$$x_2 = -1$$

$$\rightarrow \text{Hebbbar} \rightarrow \frac{(x+2)(\cancel{x+1})}{(x-3)(\cancel{x+1})}$$

$\rightarrow$ Dennoch Def. Lücke

② Asymptoten

$$\left. \begin{array}{l} \underline{\underline{x_{P1} = -1}} \\ \underline{\underline{x_{P2} = 3}} \end{array} \right\} \text{Vertikale Asymptoten}$$

$$\frac{x^2 + 3x + 2}{x^2 - 2x - 3} \rightarrow \frac{(x+2)(x+1)}{(x-3)(x+1)} \rightarrow \frac{x+2}{x-3} \rightarrow \frac{1(x-3) + 5}{(x-3)}$$

$$\rightarrow 1 + \frac{5}{x-3} \quad \xrightarrow{x \rightarrow \pm \infty} \underline{\underline{1}}$$

$$\underline{\underline{x_{P3} = 1}} \rightarrow \text{Horizontale Asymptote}$$

③ Nullstellen

$$(x+2)(x+1) = 0$$

$$\underline{\underline{x_1 = -2}}$$

$$\underline{\underline{x_2 = -1}} \rightarrow \notin \mathbb{D}$$

④ Extremstellen

$$\frac{x^2 + 3x + 2}{x^2 - 2x - 3} \rightarrow \begin{array}{ll} u(x) = x^2 + 3x + 2 & u'(x) = 2x + 3 \\ v(x) = x^2 - 2x - 3 & v'(x) = 2x - 2 \end{array}$$

$$\frac{(2x+3)(x^2-2x-3) - (x^2+3x+2)(2x-2)}{((x-3)(x+1))^2}$$

$$\frac{(2x+3)(x-3)(x+1) - (x+2)(x+1)(2x-2)}{(x-3)^2(x+1)^2}$$

$$f'(x) = \frac{(2x+3)(x-3) - (x+2)(2x-2)}{(x-3)^2(x+1)}$$

$$\rightarrow (2x+3)(x-3) - (x+2)(2x-2) \stackrel{!}{=} 0$$

$$2x^2 - 6x + 3x - 9 - (2x^2 - 2x + 4x - 4) = 0$$

$$-6x - 2x + 3x - 9 + 4 = 0$$

$$-5x = 5$$

$$\underline{x_1 = -1} \rightarrow \notin \text{II}$$

→ keine Extremstellen

⑤ Wendepunkte

$$f'(x) = \frac{(2x+3)(x-3) - (x+2)(2x-2)}{(x-3)^2(x+1)}$$

$$\frac{2x^2 - 6x + 3x - 9 - (2x^2 - 2x + 4x - 4)}{11}$$

$$\frac{-5x - 5}{(x-3)^2(x+1)}$$

$$\frac{11}{(x^2 - 6x + 9)(x+1)}$$

$$\rightarrow \frac{11}{x^3 - 6x^2 + 9x + x^2 - 6x + 9}$$

$$\rightarrow \frac{-5x-5}{x^3-5x^2+3x+9} \quad \rightarrow u(x) = -5x-5 \quad u'(x) = -5$$

$$\rightarrow v(x) = x^3-5x^2+3x+9 \quad v'(x) = 3x^2-10x+3$$

$$f''(x) = (-5x^3+25x^2-15x-45) - ($$

$$c) f(x) = \frac{x^3 - 8}{4x} \quad \rightarrow x \in \mathbb{R} \setminus \{0\}$$

① Nullstellen

$$x^3 = 8$$

$$\underline{\underline{x = 2}}$$

② Asymptoten

→ Horizontale Asymptote : $x = 0$

③ Extremstellen

$$\frac{x^3 - 8}{4x} \quad \rightarrow u(x) = x^3 - 8 \quad u'(x) = 3x^2$$

$$\rightarrow v(x) = 4x \quad v'(x) = 4$$

$$f'(x) = \frac{3x^2(4x) - (x^3 - 8)(4)}{16x^2}$$

$$= \frac{3x^3 - x^3 - 8}{4x^2}$$

$$f'(x) = \frac{x^3 - 4}{2x^2}$$

$$x^3 - 4 = 0 \quad \rightarrow \text{NS}$$

$$x^3 = 4$$

$$\underline{\underline{x = \sqrt[3]{4}}} \quad \rightarrow \text{Minima}$$

$$\underline{\underline{(\sqrt[3]{4} / f(\sqrt[3]{4}))}}$$

④ Symmetrien

→ keine

⑤ Wendepunkte

$$f'(x) = \frac{x^3 - 4}{2x^2} \quad u(x) = x^3 - 4 \quad u'(x) = 3x^2 \\ v(x) = 2x^2 \quad v'(x) = 4x$$

$$f''(x) = \frac{3x^2(2x^2) - (x^3 - 4)(4x)}{4x^4}$$

$$= \frac{6x^4 - 4x^4 - 16x}{4x^4}$$

$$= \frac{x^3 - 8}{2x^3}$$

$$x^3 - 8 = 0$$

$$\underline{\underline{x = 2}}$$

$$\rightarrow \underline{\underline{(2|0)}}$$

$$c) f(x) = \frac{x^3 - 8}{4x} \rightarrow D = \mathbb{R} \setminus \{0\}$$

1. Nullstellen:

$$x^3 - 8 = 0$$

$$x^3 = 8$$

$$x = \sqrt[3]{8}$$

$$\underline{x = 2}$$

2. Symmetrien:

$$f(x) = f(-x) \rightarrow \text{Achsensymmetrisch}$$

$$\frac{x^3 - 8}{4x} \neq \frac{(-x)^3 - 8}{-4x}$$

$$f(x) = -f(-x) \rightarrow \text{Punktsymmetrisch}$$

$$\frac{x^3 - 8}{4x} = \frac{-((-x)^3 - 8)}{4x}$$

$$\frac{x^3 - 8}{4x} \neq \frac{x^3 + 8}{4x}$$

\Rightarrow keine Symmetrien

3. Extremstellen:

$$f(x) = \frac{x^3 - 8}{4x} \quad \begin{matrix} u(x) & \rightarrow & u'(x) = 3x^2 \\ v(x) & \rightarrow & v'(x) = 4 \end{matrix} \quad \left. \vphantom{\frac{x^3 - 8}{4x}} \right\} \frac{u' \cdot v - u \cdot v'}{v^2}$$

$$f'(x) = \frac{(3x^2)(4x) - (x^3 - 8)(4)}{16x^2}$$

$$= \frac{3x^3 - x^3 - 8}{4x^2}$$

$$= \frac{2x^3 - 8}{4x^2}$$

$$\underline{f'(x) = \frac{x^3 - 4}{2x^2}}$$

$$\frac{x^3 - 4}{2x^2} = 0$$

$$x = \sqrt[3]{4}$$

$$\underline{x \approx 1.59}$$

$$f''(x) = \frac{(3x^2)(2x^2) - (x^3 - 4)(4x)}{4x^4}$$

$$= \frac{\frac{3}{2}x^4 - x^4 - 4x}{x^4}$$

$$= \frac{\frac{1}{2}x^4 - 4x}{x^4}$$

$$f''(x) = \frac{\frac{1}{2}x^3 - 4}{x^3}$$

$$= \frac{\frac{1}{2}x^3 - 4}{x^3}$$

$$f''(\sqrt[3]{4}) \approx -0.5$$

$$x_1 = \sqrt[3]{4} \rightarrow \text{Maximum lokal}$$

$$\underline{\underline{LM \approx (\sqrt[3]{4} / -0.49)}}$$

4. Wendepunkte:

$$\frac{\frac{1}{2}x^3 - 4}{x^3} = 0$$

$$\rightarrow f'(2) = 0$$

$$\begin{aligned} \frac{1}{2}x^3 &= 4 \\ x &= \sqrt[3]{8} \\ x &= 2 \end{aligned}$$

$$\rightarrow \underline{\underline{WP = (2/0)}}$$

⑤ Verhalten gegen Unendlich

$$f(x) = \frac{x^3 - 8}{4x} \rightarrow \lim_{x \rightarrow \infty}$$

$$\frac{x^3}{4x} \rightarrow \underline{y = \infty}$$

⑥ Verhalten an Asymptote

Asymptote : $x = 0$

$$\frac{x^3 - 8}{4x} \rightarrow \lim_{x \rightarrow 0} \stackrel{D}{=} < 0 \rightarrow \frac{-8}{-4x} \rightarrow (+) \infty$$

$$\frac{x^3 - 8}{4x} \rightarrow \lim_{x \rightarrow 0} \stackrel{D}{=} > 0 \rightarrow \frac{-8}{4x} \rightarrow (-) \infty$$

$$d) f(x) = \left(\frac{x-1}{x+1} \right)^2$$

$$\mathbb{D} = \mathbb{R} \setminus \{-1\}$$

① Nullstellen

$$(x-1)^2 = 0$$

$$\underline{\underline{x_1 = 1}} \rightarrow \in \mathbb{D}$$

② Ableitungen

$$f'(x) = \frac{4(x-1)}{(x+1)^3}$$

$$f''(x) = \frac{-8(x-2)}{(x+1)^4}$$

③ Extremstellen

$$4(x-1) = 0$$

$$x = 1 \rightarrow f''(1) = 0 \in \mathbb{D}$$

↳ Sattelstelle → keine Extremstellen

④ Wendepunkte

$$f''(x) = 0 \rightarrow x = 2$$

$$f(2) = \frac{1}{9} \rightarrow \underline{\underline{WP : (2 \mid \frac{1}{9})}}$$

⑤ Symmetrien

$$f(x) = f(-x)$$

$$f(x) = -f(-x)$$

$$\left(\frac{x-1}{x+1} \right)^2 \neq \left(\frac{-x-1}{-x+1} \right)^2$$

$$\left(\frac{x-1}{x+1} \right)^2 \neq - \left(\frac{-x-1}{-x+1} \right)^2$$

$$\frac{x-1}{x+1} \neq \frac{x+1}{x-1}$$

→ keine Symmetrien

⑥ Verhalten an Rand

$$\left(\frac{x-1}{x+1}\right)^2 \rightarrow \frac{x^2 - 2x + 1}{x^2 + 2x + 1} \xrightarrow{\text{"hebel -1"}} \frac{\frac{1}{x^2} - \frac{1}{2x} + 1}{\frac{1}{x^2} + \frac{1}{2x} + 1}$$

$$\hookrightarrow \lim_{x \rightarrow \pm\infty} \rightarrow \underline{\underline{1}}$$

⑦ Verhalten um Definitionslücke "-1"

$$\left(\frac{x-1}{x+1}\right)^2 \lim_{x \rightarrow -1} < -1 \quad \left(\frac{-1,999}{0,001}\right)^2 \rightarrow \underline{\underline{\infty}}$$

$$\left(\frac{x-1}{x+1}\right)^2 \lim_{x \rightarrow -1} > -1 \quad \left(\frac{-2,001}{-0,001}\right)^2 \rightarrow \underline{\underline{\infty}}$$

Kettenregel:

$$f(x) = u(v(x))$$

$$f'(x) = u'(v(x)) \cdot v'(x)$$

Beispiel:

$$f(x) = 3 \cdot (x^3 - 4)^7$$

$$\rightarrow u(x) = 3(x)^7 \quad \rightarrow f'(x) = u'(v(x)) \cdot v'(x)$$

$$v(x) = x^3 - 4$$

$$\hookrightarrow f'(x) = 3 \cdot 7(x^3 - 4)^6 \cdot 3x^2$$

$$\underline{\underline{f'(x) = 63x^2(x^3 - 4)^6}}$$

Leite mit der Kettenregel ab:

$$\text{Kettenregel: } f'(x) = u'(v(x)) \cdot v'(x)$$

15. a) $f(x) = (1 + \sqrt{x})^2$

$$f'(x) = (1 + \sqrt{x})^2$$

$$u(x) = x^2$$

$$u'(x) = 2x$$

$$v(x) = 1 + \sqrt{x}$$

$$v'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(x) = u'(v(x)) \cdot v'(x)$$

$$= 2(1 + \sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$$

$$= \frac{2}{2\sqrt{x}} (1 + \sqrt{x}) = \frac{2 + 2\sqrt{x}}{2\sqrt{x}}$$

$$= \frac{1 + \sqrt{x}}{\sqrt{x}} = \frac{1}{\sqrt{x}} + \frac{\sqrt{x}}{\sqrt{x}} = \underline{\underline{\frac{1}{\sqrt{x}} + 1}}$$

b) $f(x) = (2\sqrt{x} - x)^3$

$$u(x) = x^3 \quad u'(x) = 3x^2$$

$$v(x) = 2\sqrt{x} - x \quad v'(x) = \frac{1}{\sqrt{x}} - 1$$

$$f'(x) = 3(2\sqrt{x} - x)^2 \cdot \left(\frac{1}{\sqrt{x}} - 1\right)$$

$$= 3(4x - 4x\sqrt{x} + x^2) \left(\frac{1}{\sqrt{x}} - 1\right)$$

$$= 3 \left(\frac{4x}{\sqrt{x}} - 4x - \frac{4x\sqrt{x}}{\sqrt{x}} + 4x\sqrt{x} + \frac{x^2}{\sqrt{x}} - x^2 \right)$$

$$= 3 \left(\frac{4x + x^2}{\sqrt{x}} - 8x + 4x\sqrt{x} - x^2 \right)$$

$$c) f(x) = 2(x^2 - 3\sqrt{x})^2$$

$$u(x) = 2x^2$$

$$u'(x) = 4x$$

$$v(x) = x^2 - 3\sqrt{x}$$

$$v'(x) = 2x - \frac{3}{2\sqrt{x}}$$

$$f'(x) = 4(x^2 - 3\sqrt{x}) \left(2x - \frac{3}{2\sqrt{x}} \right)$$

$$= 4 \left(2x^3 - \frac{3x^2}{2\sqrt{x}} + \frac{9\sqrt{x}}{2\sqrt{x}} - 6x\sqrt{x} \right)$$

$$= 4 \left(2x^3 - \frac{9}{2} - \frac{3x^2}{2\sqrt{x}} - 6x\sqrt{x} \right) \quad 8x^3 - 30x^{1.5} + 18$$

$$= 8x^3 - 18 - \frac{3x^2}{2x^{0.5}} - 6x^2 \cdot x^{0.5}$$

$$= 8x^3 - 18 - 6x^{1.5} - 24x^{1.5}$$

$$= \underline{\underline{8x^3 - 30x^{1.5} + 18}}$$

$$d) g(t) = (t^3 - \sqrt{t})^{-2}$$

$$u(x) = x^{-2}$$

$$u'(x) = -2x^{-3}$$

$$v(x) = (t^3 - \sqrt{t}) \quad v'(x) = 3t^2 - \frac{1}{2\sqrt{t}}$$

$$f'(x) = -2(t^3 - \sqrt{t})^{-3} \cdot \left(3t^2 - \frac{1}{2\sqrt{t}} \right)$$

$$f'(x) = \underline{\underline{\frac{-2 \left(3t^2 - \frac{1}{2\sqrt{t}} \right)}{(t^3 - \sqrt{t})^3}}}$$

$$e) f(x) = (\sin(x))^2 = \sin^2(x)$$

$$u(x) = x^2$$

$$u'(x) = 2x$$

$$v(x) = \sin(x)$$

$$v'(x) = \cos(x)$$

$$f'(x) = 2(\sin(x)) \cdot \cos(x)$$

$$\underline{\underline{f'(x) = \sin(2x)}}$$

$$f) g(x) = (\cos(x))^3 = \cos^3(x)$$

$$u(x) = x^3 \quad u'(x) = 3x^2$$

$$v(x) = \cos(x) \quad v'(x) = -\sin(x)$$

$$f'(x) = 3(\cos(x))^2 \cdot -\sin(x)$$

$$\underline{\underline{f'(x) = -3\sin(x) \cdot \cos^2(x)}}$$

$$17. a) f(x) = \frac{(2x+1)^2}{3x-5}$$

$$\rightarrow g(x) = (2x+1)^2$$

$$\rightarrow u(x) = x^2 \quad u'(x) = 2x$$

$$v(x) = 2x+1 \quad v'(x) = 2$$

$$\rightarrow 2(2x+1) \cdot 2 \rightarrow \underline{8x+4} = g'(x) = u'(x)$$

$$f(x) = \frac{(2x+1)^2}{3x-5} \rightarrow \frac{u'v - uv'}{v^2}$$

$$f(x) = \frac{(8x+4)(3x-5) - (2x+1)^2(2)}{(3x-5)^2}$$

$$= \frac{24x^2 + 12x - 40x - 20 - 12x^2 - 12x - 2}{(3x-5)^2}$$

$$= \frac{12x^2 - 40x - 23}{(3x-5)^2}$$

$$\underline{\underline{\hspace{10cm}}}$$

$$b) f(x) = \frac{1+2x}{(2-x)^2}$$

$$\rightarrow g(x) = (2-x)^2$$

$$u(x) = x^2 \quad u'(x) = 2x$$

$$v(x) = 2-x \quad v'(x) = -1$$

$$g'(x) = -2x(2-x) = 2x^2 - 4x$$

$$f(x) = \frac{2(2-x)^2 - (1+2x)(2x^2 - 4x)}{(2-x)^4} \rightarrow \frac{2(2-x)^2 - (1+2x) \cdot -2x(2-x)}{(2-x)^4}$$

$$\rightarrow \frac{2(2-x) - (1+2x)(-2x)}{(2-x)^3} \rightarrow \frac{-2x+4+2x+4x^2}{(2-x)^3}$$

$$\rightarrow \frac{4x^2+4}{(2-x)^3}$$

$$c) f(x) = \frac{(1-2x)^2}{(3+5x)^2}$$

$$\text{LR: } f'(x) = u'(v(x)) \cdot v'(x)$$

$$\begin{aligned} u(x) &= x^2 & u'(x) &= 2x \\ v(x) &= (1-2x) & v'(x) &= -2 \end{aligned} \quad \rightarrow \quad \begin{aligned} &2(1-2x) \cdot (-2) \\ &\hookrightarrow -4+8x \end{aligned}$$

$$\begin{aligned} u &= x^2 & u' &= 2x \\ v &= 3+5x & v' &= 5 \end{aligned} \quad \rightarrow \quad \begin{aligned} &2(3+5x) \cdot 5 \\ &\hookrightarrow 30+50x \end{aligned}$$

$$\begin{aligned} u(x) &= (1-2x)^2 & u'(x) &= -4+8x \\ v(x) &= (3+5x)^2 & v'(x) &= 30+50x \end{aligned} \quad \rightarrow \quad \frac{u'v - uv'}{v^2}$$

$$\frac{(-4+8x)(3+5x)^2 - (1-2x)^2(30+50x)}{(3+5x)^4}$$

$$\frac{(-4+8x)(3+5x) - (1-2x)^2(10)}{(3+5x)^3}$$

$$\frac{(-12 + 24x - 20x + 40x^2) - (10 - 40x + 40x^2)}{(3+5x)^3}$$

$$\frac{-22 + 44x}{(3+5x)^3}$$

$$d) f(x) = \left(\frac{3x+2}{4x-2} \right)^2$$

$$\rightarrow \begin{array}{ll} u = 3x+2 & u' = 3 \\ v = 4x-2 & v' = 4 \end{array} \rightarrow \frac{u'v - uv'}{v^2}$$

$$\rightarrow \frac{3(4x-2) - 4(3x+2)}{(4x-2)^2} \rightarrow \frac{(12x-6) - (12x+8)}{(4x-2)^2}$$

$$\rightarrow \frac{-14}{(4x-2)^2}$$

$$u(x) = x^2 \quad u'(x) = 2 \quad \rightarrow u'(v(x)) \cdot v'(x)$$

$$v(x) = \frac{3x+2}{4x-2} \quad v'(x) = \frac{-14}{(4x-2)^2}$$

$$\rightarrow 2 \left(\frac{3x+2}{4x-2} \right) \left(\frac{-14}{(4x-2)^2} \right) \rightarrow \frac{(6x+4) - 14}{(4x-2)(4x-2)^2}$$

$$\frac{-28(3x+4)}{2(2x-1) \cdot 2(x-1) \cdot 2(x-1)} \sim \frac{-7(3x+4)}{2(2x-1)^3}$$

$$i) f(t) = \frac{(2t+1)^2}{\sqrt{1-t}}$$

$$\rightarrow 4t^2 + 4t + 1 \rightarrow \underline{u'} = 8t + 4$$

$$\rightarrow u = \sqrt{x} \quad u' = \frac{1}{2\sqrt{x}} \rightarrow \underline{v'} = \frac{-1}{2\sqrt{1-t}}$$

$$v = 1-t \quad v' = -1$$

$$\rightarrow f'(t) = \frac{(8t+4)(\sqrt{1-t}) - (2t+1)^2 \left(\frac{-1}{2\sqrt{1-t}} \right)}{(1-t)}$$

$$\frac{(8t+4)(\sqrt{1-t}) - (2t+1)^2 (2\sqrt{1-t})^{-1} (-1)}{(1-t)}$$

$$\frac{(8t+4)(1-t) - (2t+1)^2 \left(\frac{1}{2} \right) (-1)}{(1-t)^{\frac{3}{2}}}$$

$$\frac{(8t - 8t^2 + 4 - 4t) + (2t^2 + 2t + \frac{1}{2})}{(1-t)^{\frac{3}{2}}}$$

$$-6t^2 + 6t + 4.5$$

$$\frac{-6t^2 + 6t + 4.5}{(1-t)^{\frac{3}{2}}}$$

